



ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY FOR INDUSTRY

| COLLEGE OF ENGINEERING.

| DEPARTMENT OF SOFTWARE ENGINEERING.

| FUNDAMENTALS OF PROGRAMMING II (SwEg2103).

PROJECT REPORT

TITLE: SMART STAFF ATTENDANCE AND PAYROLL
MANAGEMENT SYSTEM.

| SEC: B

Name	I.D. NUMBER
1: HUNDAOL BEGNA	ETS0756/17
2: ISAAC ASSEFA	ETS0766/17
3: GEMMACHU FIKADU	ETS0628/17
4: IFA SILASHI	ETS0764/17
5: GURMESA BORENA	ETS0661/17

Submitted To: Eleni Teshome

Submission Date: May-15-2026

Background

Organizations and institutions often face challenges in managing employee attendance and payroll operations manually. Traditional payroll systems require significant human effort, consume time, and are prone to calculation errors. Manual attendance tracking also creates problems in maintaining accurate employee records.

To address these issues, the Smart Staff Attendance & Payroll Management System was developed using C++ and MySQL. The system automates employee management, attendance recording, payroll calculation, and administrative control.

The project demonstrates the integration of object-oriented programming concepts with relational database management systems to build a real-world software solution.

Problem Statement

Many organizations still rely on manual methods for managing employee attendance and salary processing. These methods introduce several challenges including:

- Payroll calculation errors
- Poor attendance tracking
- Data redundancy
- Lack of security
- Difficulty managing employee records
- Time-consuming administrative tasks

The absence of automation reduces efficiency and increases the possibility of human mistakes.

Objectives

General Objective

To develop a secure and efficient payroll management system using C++ and MySQL.

Specific Objectives

- To automate attendance management
- To automate payroll calculation
- To implement staff management operations
- To integrate MySQL database connectivity
- To provide role-based authentication
- To improve data organization and integrity

Scope of the System

The system focuses on:

- Employee information management
- Attendance recording
- Payroll generation
- Administrative login control
- Database-driven data storage

The project does not include:

- Online deployment
- Biometric attendance
- Mobile application integration
- Cloud synchronization

System Analysis

Existing System Analysis

Traditional payroll systems commonly use paper-based records or spreadsheet applications. These systems are inefficient because they:

- Require repetitive manual calculations
- Lack centralized data management
- Are difficult to scale
- Have weak security mechanisms
- Increase administrative workload

Proposed System

The proposed Smart Staff Attendance & Payroll Management System automates attendance and payroll processing using a structured software architecture.

The system provides:

- Automated salary calculation
- Attendance monitoring
- Employee management
- Authentication system
- Database integration
- Administrative control

Feasibility Analysis

Technical Feasibility

The project is technically feasible because the required tools and technologies are available and accessible.

Economic Feasibility

The system minimizes paperwork and reduces operational inefficiencies.

Operational Feasibility

The system is user-friendly and suitable for organizations requiring attendance and payroll management.

Functional and Non-Functional Requirements

Functional Requirements

The system shall:

- Allow admin login
- Create users
- Add staff records
- Update employee information
- Delete staff records
- Record attendance
- Generate payroll
- Search employee records
- Display attendance reports

Non-Functional Requirements

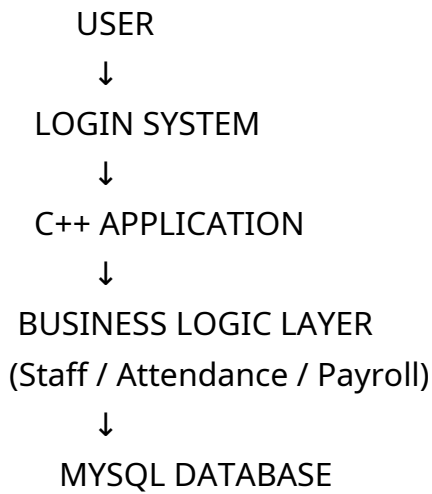
The system shall provide:

- Security
- Reliability
- Data consistency
- Efficiency
- Maintainability
- Scalability

System Design

System Architecture

The system follows a modular architecture.



UML Class Design

The project contains the following major classes:

Auth Class

Handles authentication and user creation.

Staff Class

Handles employee management operations.

Attendance Class

Handles attendance recording and monitoring.

Payroll Class

Handles payroll calculation and reporting.

Entity Relationship Design

Entities used:

- Users
- Staff
- Attendance
- Payroll

Relationships:

- One staff member can have multiple attendance records.
- One staff member can have multiple payroll records.

Database Design

Users Table

Field	Type
user_id	INT
username	VARCHAR
password	VARCHAR
role	VARCHAR

Staff Table

Field	Type
staff_id	INT
name	VARCHAR
department	VARCHAR
salary_per_day	DOUBLE

Attendance Table

Field	Type
attendance_id	INT
staff_id	INT
month	VARCHAR
days_present	INT

Payroll Table

Field	Type
payroll_id	INT
staff_id	INT
month	VARCHAR
total_salary	DOUBLE
bonus	DOUBLE
deduction	DOUBLE

Implementation

Technologies Used

Technology	Purpose
C++	Application logic
MySQL	Database management
MySQL Connector/C++	Database connectivity
Git & GitHub	Version control
Linux Mint	Development environment

Authentication Module

The authentication module validates user credentials before granting access to the system.

Features:

- Admin login
- User creation
- Role-based access concept

Staff Management Module

The staff module supports:

- Adding staff
- Viewing staff
- Updating staff records

- Deleting staff
- Searching staff

Attendance Module

The attendance module allows administrators to:

- Record attendance
- View attendance
- Detect low attendance

Payroll Module

The payroll module calculates:

- Total salary
- Bonus
- Deduction
- Final payroll output

Testing

Login Testing

The login system was tested using valid and invalid credentials.

Result:

- Valid users successfully accessed the system.
- Invalid users were denied access.

Staff Module Testing

The system successfully:

- Added staff
- Updated records
- Deleted records
- Searched staff information

Attendance Testing

Attendance data was successfully recorded and retrieved from the database.

Payroll Testing

Payroll calculations were tested using attendance records and salary information.

The system correctly generated payroll outputs.

Challenges Faced

During development, several challenges were encountered:

- MySQL connection errors
- Foreign key constraint issues
- File structure organization
- GitHub integration challenges
- Authentication integration issues

These challenges were solved through debugging, database restructuring, and modular code organization.

Future Improvements

Future upgrades may include:

- GUI/Desktop interface
- Password encryption
- PDF payroll reports
- CSV export functionality
- Web deployment
- Analytics dashboard
- Cloud database integration

Conclusion

The Smart Staff Attendance & Payroll Management System successfully demonstrates the integration of C++ and MySQL to solve real-world payroll and attendance management problems.

The system improved efficiency, reduced manual work, and provided secure database-driven operations.

The project also strengthened understanding of:

- Object-oriented programming
- Database systems
- Authentication systems
- Software architecture
- Version control using GitHub

References

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2. MySQL Documentation — <https://dev.mysql.com/doc/>
3. GitHub Documentation — <https://docs.github.com/>
4. MySQL Connector/C++ Documentation
5. Software Engineering Course Materials